

Muhammad Abdullah

Backend Developer FastAPI

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PROFESSIONAL SUMMARY

BS Software Engineering graduate specializing in backend development with FastAPI, backed by a strong foundation in AI, machine learning, and NLP. Experienced in building production grade backend systems with JWT authentication, PostgreSQL/SQLAlchemy data layers, Alembic migrations, asynchronous task pipelines (Celery/Redis), and AI-powered features via LLM API integration (OpenRouter, Groq, Gemini). Fine-tuned OpenAI Whisper (0.8B) for Urdu ASR, publishing 9 model checkpoints on Hugging Face, and built full-stack AI web apps with Streamlit, Flask, and React. Proficient in Python and modern backend/AI tooling, with hands-on data, BI, and containerized deployment experience across five industry internships.

TECHNICAL SKILLS

Backend & APIs: FastAPI, REST API Design, Pydantic, JWT Authentication, SQLAlchemy (ORM), Alembic (Migrations), Uvicorn, Swagger

Languages: Python, SQL, C++, HTML/CSS

Databases: PostgreSQL, MySQL, XAMPP

AI / ML: Model Fine-Tuning, NLP, Hugging Face Transformers, OpenAI Whisper, BART, Scikit-learn, LLM API Integration

Background Tasks & Messaging: Celery, Redis

DevOps & Tools: Docker, Docker Compose, Git/GitHub, GitHub Actions (CI/CD), Postman, Streamlit, Flask

Platforms: Hugging Face Hub, Kaggle, Google Colab, Microsoft Power BI

EXPERIENCE

Backend Developer

Jun 2026 – Present

Whooplt | Digital Marketing Solutions

- Developed and maintained backend services across multiple client projects using Python and FastAPI.
- Designed and queried PostgreSQL data models to support core product features and business logic.
- Contributing to a production codebase as part of a cross-functional engineering team.

Lead Generation Specialist

Nov 2024 – Jan 2025

Silverlight Research (Remote)

- Generated high-quality B2B leads via LinkedIn, targeting decision-makers and senior industry professionals.
- Leveraged Apollo.io and Exa to identify, verify, and extract accurate contact and company data.
- Built structured lead lists supporting outreach campaigns and improved quality through targeted filtering.

Multi-Sector Intern

Jul – Sept 2024

NADRA Headquarters, Islamabad

- Supported identity verification and data management workflows under strict security and privacy compliance.
- Worked with enterprise-scale databases, gaining practical exposure to data validation and information systems.
- Assisted in automating data processing tasks, improving operational throughput and accuracy.

Python Developer Intern

Jul 2024

CodexCue (Remote)

- Developed Python automation scripts to eliminate repetitive tasks and improve workflow efficiency.
- Debugged and optimized existing codebase, improving application performance and reliability.

Business Intelligence Intern

Jul – Aug 2023

PTCL Headquarters, Islamabad

- Managed and queried relational databases in MySQL to support data analysis and reporting.
- Designed interactive Power BI dashboards translating raw data into actionable business insights.
- Created Entity-Relationship Diagrams (ERDs) for data modeling and system documentation.

- Gained practical knowledge of telecom network infrastructure, monitoring, and operational workflows.
- Developed Python scripts for data handling and automation across network operations.
- Generated network performance reports using Power BI, assisting in identifying performance bottlenecks.

PROJECTS

Handover Voice Summarizer AI-Powered Care Home Staff Management System

FastAPI · PostgreSQL · React · Celery · Redis · Docker · Whisper · Gemini [GitHub ↗](#)

- Built a full-stack care home staff management system using FastAPI, PostgreSQL, and React to digitize shift handover reporting between care workers and managers.
- Implemented local audio transcription with OpenAI Whisper and AI-powered summarization with Gemini to convert voice notes into structured reports (key events, medications, incidents, urgency level).
- Architected an asynchronous processing pipeline using Celery and Redis for non-blocking transcription and summarization, with a shared Docker volume for cross-container audio access.
- Designed secure JWT-based authentication with OTP email verification (Resend) and rate limiting (slowapi) to prevent abuse.
- Containerized a 5-service architecture (API, Celery worker, Celery beat, PostgreSQL, Redis, Nginx-served frontend) using Docker Compose for consistent local and production environments.
- Set up a GitHub Actions CI/CD pipeline with automated pytest testing and Alembic migrations against a live PostgreSQL service container.

Job Tracker AI-Powered Job Application Backend

FastAPI · PostgreSQL · SQLAlchemy · Alembic · JWT [GitHub ↗](#)

- Built a production-style REST API for tracking job applications, with JWT-based authentication and role-secured endpoints.
- Designed a relational schema with SQLAlchemy and versioned it using Alembic migrations to support scalable multi-user data.
- Integrated Gemini model to power an AI resume-analysis feature that score resumes against job descriptions and returns actionable feedback.
- Developed "Pathway," a companion frontend client consuming the API end-to-end for a complete user experience.

Thumbnail Generator AI Thumbnail Creation API

FastAPI · JWT · Pollinations.ai · Pydantic · SQLAlchemy [GitHub ↗](#)

- Built a FastAPI backend enabling authenticated users to generate AI-powered YouTube thumbnails, integrating Pollinations.ai for image generation with dynamic title-text overlay.
- Implemented JWT authentication and full CRUD endpoints for creating, saving, and managing generated thumbnails.
- Designed request/response schemas with Pydantic validation and auto-generated OpenAPI/Swagger documentation.

Social Platform API

FastAPI · PostgreSQL · SQLAlchemy · JWT · Docker [GitHub ↗](#)

- Developed a full-stack social platform API (posts, votes, users) with FastAPI, PostgreSQL, and SQLAlchemy, including relational vote-tracking logic and JWT-secured routes.
- Built a complete vanilla JavaScript single-page frontend (no build step) consuming the API end-to-end.
- Containerized the app with Docker and deployed it on Hugging Face Spaces for public access.

Video Analysis System Final Year Project

Python · OpenAI Whisper · BART · NLP · Streamlit [Hugging Face ↗](#)

- Built a full-stack web app for end-to-end video transcription, summarization, and content management.
- Fine-tuned OpenAI Whisper (0.8B) on custom Urdu datasets and published 9 model checkpoints on Hugging Face.
- Implemented NLP summarization with BART and subtitle keyword search with segment editing/deletion.

EDUCATION

Bachelor of Science in Software Engineering

Oct 2021 – Jun 2025

Pak-Austria Fachhochschule: Institute of Applied Sciences and Technology (PAF-IAST), Haripur